

## E. G. S. PILLAY ENGINEERING COLLEGE

(Autonomous)

Approved by AICTE, New Delhi| Affiliated to Anna University, Chennai Accredited by

NAAC with A++ Grade| Accredited by NBA T1

NAGAPATTINAM–611002



### B.E. Biomedical Engineering

Final Year – Eighth Semester

| COURSE<br>CODE    | COURSENAME              | CATEGORY | L | T | P | C | MAX.MARKS |    |       |
|-------------------|-------------------------|----------|---|---|---|---|-----------|----|-------|
|                   |                         |          |   |   |   |   | CA        | ES | TOTAL |
| PRACTICAL COURSES |                         |          |   |   |   |   |           |    |       |
| 2302BM851         | Internship Presentation | EEC      | 0 | 0 | 0 | 1 | 100       | -  | 100   |
| 2302BM852         | Project Work            | EEC      | 0 | 0 |   | 8 | 50        | 50 | 100   |
| TOTAL             |                         |          |   |   |   | 9 | 150       | 50 | 200   |

L–Lecture |T–Tutorial | P–Practical| C–Credit| CA–Continuous Assessment |ES–End Semester

|                                                                                                                       |                         |     |     |     |     |     |     |     |     |      |      |      |      |      |   |
|-----------------------------------------------------------------------------------------------------------------------|-------------------------|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|---|
| 2302BM851                                                                                                             | INTERNSHIP PRESENTATION |     |     |     |     |     |     |     |     |      |      | L    | T    | P    | C |
|                                                                                                                       |                         |     |     |     |     |     |     |     |     |      |      | 0    | 0    | 0    | 1 |
| COURSE OBJECTIVES:                                                                                                    |                         |     |     |     |     |     |     |     |     |      |      |      |      |      |   |
| The student should be made to:                                                                                        |                         |     |     |     |     |     |     |     |     |      |      |      |      |      |   |
| 1. Provide hands-on experience at site where biomedical equipment are manufactured and utilized (Hospitals).          |                         |     |     |     |     |     |     |     |     |      |      |      |      |      |   |
| COURSE OUTCOME:                                                                                                       |                         |     |     |     |     |     |     |     |     |      |      |      |      |      |   |
| 1.Students have to undergo two weeks practical training in biomedical equipment manufacturing companies or hospitals. |                         |     |     |     |     |     |     |     |     |      |      |      |      |      |   |
| 2.At the end of the training student will submit a report as per the prescribed format to the department              |                         |     |     |     |     |     |     |     |     |      |      |      |      |      |   |
| COs Vs POs and PSO MAPPING:                                                                                           |                         |     |     |     |     |     |     |     |     |      |      |      |      |      |   |
|                                                                                                                       |                         |     |     |     |     |     |     |     |     |      |      |      |      |      |   |
| CO No                                                                                                                 | PO1                     | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 | PSO3 |   |
| 1                                                                                                                     | 2                       | 2   | 3   | 3   | 1   | 3   | 3   | 3   | 3   | 3    |      | 3    | 3    | 3    |   |
| 2                                                                                                                     | 2                       | 2   | 3   | 3   | 1   | 3   | 3   | 3   | 3   | 3    |      | 3    | 3    | 3    |   |
| 3                                                                                                                     | 2                       | 2   | 3   | 3   | 3   | 3   | 3   | 3   | 3   | 3    | 2    | 3    | 3    | 3    |   |
| 4                                                                                                                     | 2                       | 2   | 3   | 3   | 3   | 3   | 3   | 3   | 3   | 3    | 2    | 3    | 3    | 3    |   |
| 5                                                                                                                     | 2                       | 2   |     |     | 2   |     |     | 3   | 3   | 3    |      | 3    | 3    | 3    |   |

|                                                         |                                                                                                                                                          |     |     |     |     |     |     |     |     |      |      |      |      |      |   |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|---|
| 2302BM852                                               | PROJECT WORK                                                                                                                                             |     |     |     |     |     |     |     |     |      |      | L    | T    | P    | C |
|                                                         |                                                                                                                                                          |     |     |     |     |     |     |     |     |      |      | 0    | 0    |      | 8 |
| COURSE OBJECTIVES:                                      |                                                                                                                                                          |     |     |     |     |     |     |     |     |      |      |      |      |      |   |
|                                                         | 1.Work in teams to propose, formulate, and solve a challenging open-ended design problem of significant scope, depth, and breadth.                       |     |     |     |     |     |     |     |     |      |      |      |      |      |   |
|                                                         | 2.Understand and incorporate engineering standards and multiple realistic constraints, within realistic design time, budget, and performance objectives. |     |     |     |     |     |     |     |     |      |      |      |      |      |   |
|                                                         | 3.Develop a prototype of the proposed design and demonstrate the prototype in accordance with the specifications.                                        |     |     |     |     |     |     |     |     |      |      |      |      |      |   |
|                                                         | 4.Effectively communicate information relating to all aspects of the design process in written, oral, and graphical form.                                |     |     |     |     |     |     |     |     |      |      |      |      |      |   |
|                                                         |                                                                                                                                                          |     |     |     |     |     |     |     |     |      |      |      |      |      |   |
| COURSE OUTCOMES:                                        |                                                                                                                                                          |     |     |     |     |     |     |     |     |      |      |      |      |      |   |
| After completion of the course, Student will be able to |                                                                                                                                                          |     |     |     |     |     |     |     |     |      |      |      |      |      |   |
| CO1                                                     | Formulate a real world problem, identify the requirement and develop the design solutions.                                                               |     |     |     |     |     |     |     |     |      |      |      |      |      |   |
| CO2                                                     | Identify technical ideas, strategies and methodologies.                                                                                                  |     |     |     |     |     |     |     |     |      |      |      |      |      |   |
| CO3                                                     | Utilize the new tools, algorithms, techniques that contribute to obtain the solution of the project.                                                     |     |     |     |     |     |     |     |     |      |      |      |      |      |   |
| CO4                                                     | Test and validate through conformance of the developed prototype and analysis the cost effectiveness.                                                    |     |     |     |     |     |     |     |     |      |      |      |      |      |   |
| CO5                                                     | Prepare a report and present the oral demonstrations.                                                                                                    |     |     |     |     |     |     |     |     |      |      |      |      |      |   |
|                                                         |                                                                                                                                                          |     |     |     |     |     |     |     |     |      |      |      |      |      |   |
| COs Vs POs and PSO MAPPING:                             |                                                                                                                                                          |     |     |     |     |     |     |     |     |      |      |      |      |      |   |
|                                                         |                                                                                                                                                          |     |     |     |     |     |     |     |     |      |      |      |      |      |   |
| CO No                                                   | PO1                                                                                                                                                      | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 | PSO3 |   |
| 1                                                       | 2                                                                                                                                                        | 2   | 3   | 3   | 1   | 3   | 3   | 3   | 3   | 3    |      | 3    | 3    | 3    |   |
| 2                                                       | 2                                                                                                                                                        | 2   | 3   | 3   | 1   | 3   | 3   | 3   | 3   | 3    |      | 3    | 3    | 3    |   |
| 3                                                       | 2                                                                                                                                                        | 2   | 3   | 3   | 3   | 3   | 3   | 3   | 3   | 3    | 2    | 3    | 3    | 3    |   |
| 4                                                       | 2                                                                                                                                                        | 2   | 3   | 3   | 3   | 3   | 3   | 3   | 3   | 3    | 2    | 3    | 3    | 3    |   |
| 5                                                       | 2                                                                                                                                                        | 2   |     |     | 2   |     |     | 3   | 3   | 3    |      | 3    | 3    | 3    |   |